

P10 Presentation

Team: t05

The Code Monkeys

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"We are striving to write high quality code and helping each other improve while creating and developing successful products"



Redesign

What has changed

- Cohesion & Coupling
 - Domain model
 - Class model (UML)
-

Redesign: Cohesion & Coupling

We found a couple things that violated the **Single Responsibility Principle (SRP)** along with some instances of high stamp coupling:

Class	Coupling	Cohesion	Comments
User	Stamp, Data	Communicational, Informational, Functional	High stamp coupling, violates SRP
Rules	Stamp, Data	Functional, Communicational	High stamp coupling, violates SRP
Invitation	Data	Functional	---
Profile	Stamp, Data	Functional, Informational	---
Match	Stamp, Data	Logical, Functional, Communicational	High Stamp Coupling
Game	Data, Common	Functional, Temporal	---
Notification	Data	Communicational	---
Platform	Stamp, Data	Functional	---
History	Stamp	Functional	---
Piece	Data	Functional	---

Actions taken:

- **Defined interfaces** for Notification and Invitation
 - *reduce direct dependencies*
- **Split large classes** into SRP compliant classes
 - Ex: *Rules.java* and *Piece.java*
- **Set and passed unique user IDs** instead of entire User Class
 - *to help with Stamp and Data coupling*

Redesign: Domain model (CRC Cards)

CRC Card	Owner	Notes
Platform	Dureke	N/A
User	John	N/A
Match	Ethan	N/A
Invitation	Matthew	N/A
Notification	Samuel	N/A
History	Dureke	N/A
Rules	John	N/A
State	Ethan	deprecated
Profile	Matthew	N/A
Game	Samuel	N/A

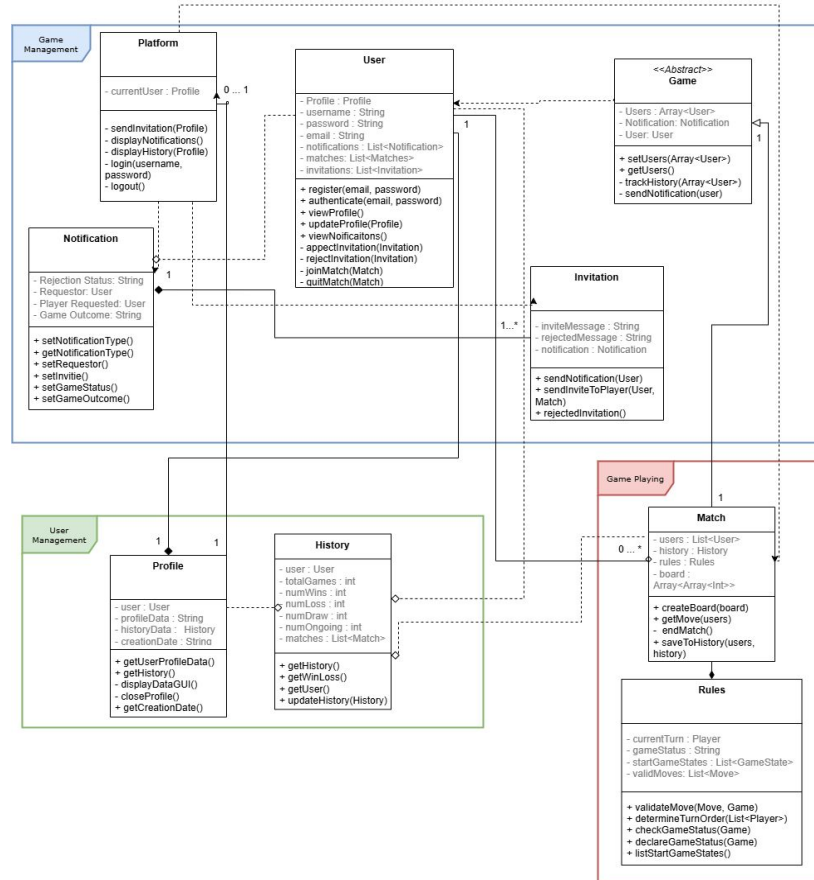


CRC Card	Owner	Notes
Credential	Dureke	N/A
Database	John	N/A
Game	Ethan	N/A
GameStatus	Matthew	N/A
Invitation	Samuel	N/A
Notification	Dureke	N/A
Platform	John	N/A
User	Ethan	N/A
Bishop	Matthew	N/A
Color	Samuel	N/A
King	Dureke	N/A
Knight	John	N/A
Match	Ethan	N/A
Move	Matthew	N/A
Pawn	Samuel	N/A
Piece	Dureke	N/A
PieceFactory	John	N/A
PieceType	Ethan	N/A
Queen	Matthew	N/A
Rook	Samuel	N/A
Rules	Dureke	N/A
History	John	N/A
Profile	Ethan	N/A

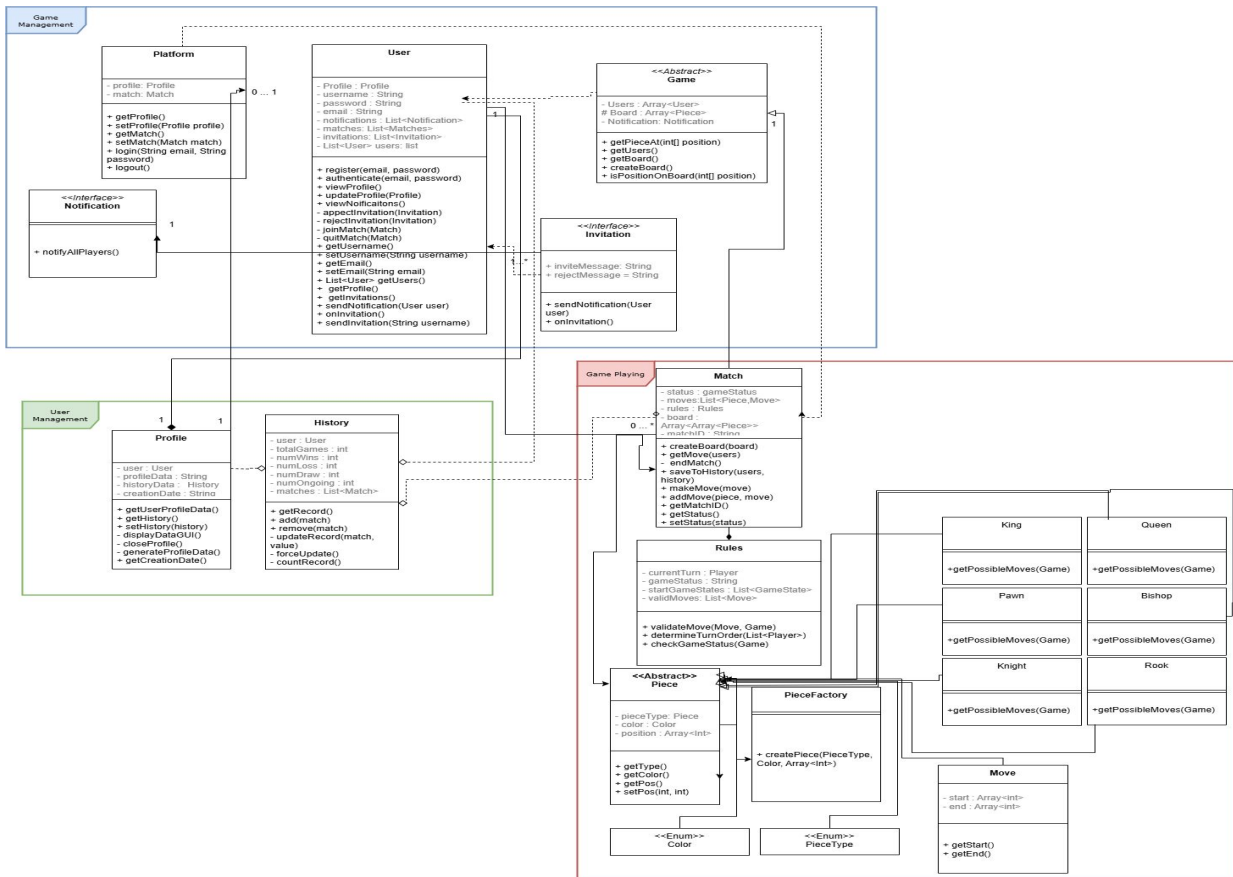
Throughout this sprint we have updated our original **CRC cards table**

These have gone from having only basic domain classes to new specific classes, like *Bishop*, *Move*, *PieceFactory*, etc.

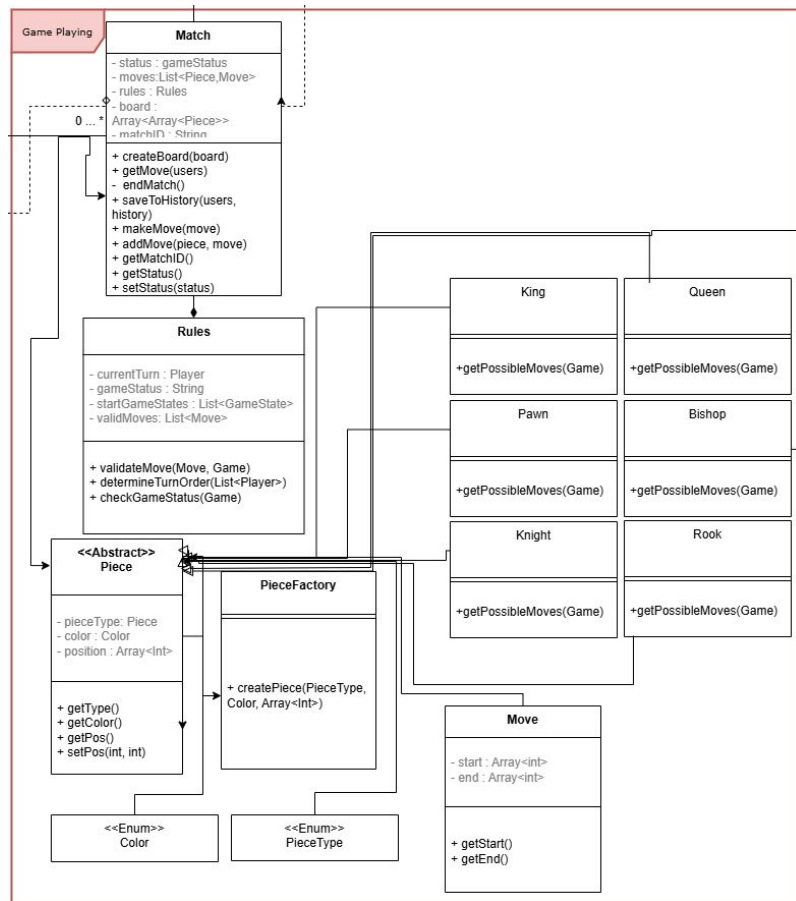
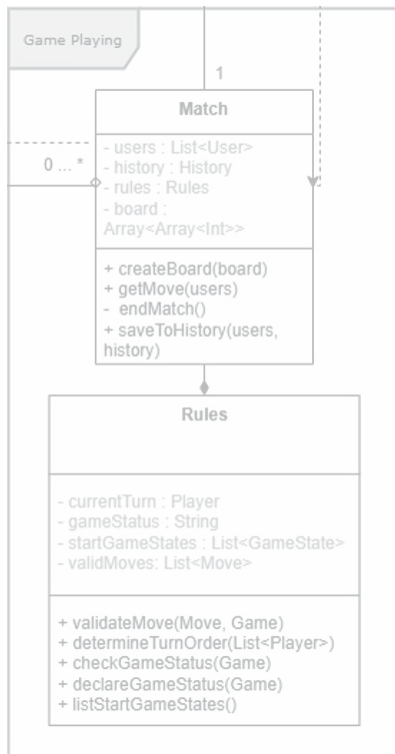
Redesign: Class model (UML) Before



Redesign: Class model (UML) **After**



Piece Classes



J Bishop.java
 J Color.java
 J King.java
 J Knight.java
 J Match.java
 J Move.java
 J Pawn.java
 J Piece.java
 J PieceFactory.java
 J PieceType.java
 J Queen.java
 J Rook.java
 J Rules.java

Implementation Progress

Server code:

- Pieces

- Added new Piecefactory class with supporting classes: Color, PieceType, and Piece
- Every piece is now its own class and extends Piece (i.e. Rook, Pawn, King)

- Rules

- Implemented important game rules like turnOrder and checkGameStatus to Rules class
- Added a piece's potential/ allowed moves to their respective classes

- Match

- Added unique match UUIDs
- Added and implemented supporting Move class
- Added functionality to allow moves to be made if validated using

- Game

- Added and implemented supporting GameState class
- Added functionality for creating the board and locating pieces

- History

- Added and implemented supporting GameState class

- Platform

- Added new login functionality

- Database

- Added 2 new classes Credential, and Database to send and receive match data

- Notification & Invitation

- Changed to interfaces

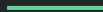
Client UI Development

- Work in Progress!



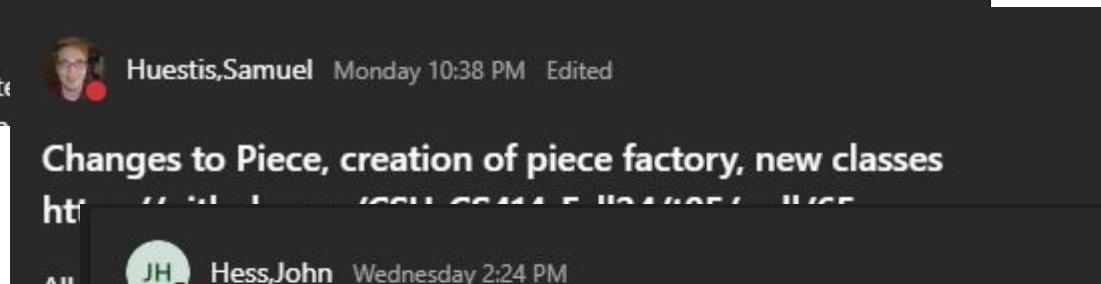
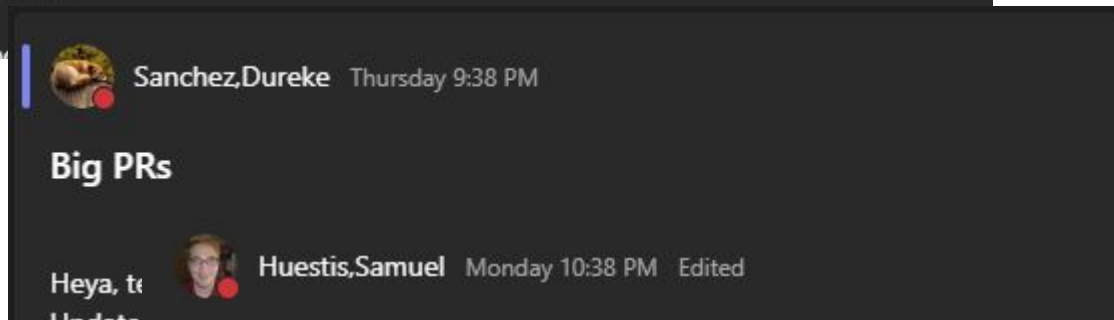
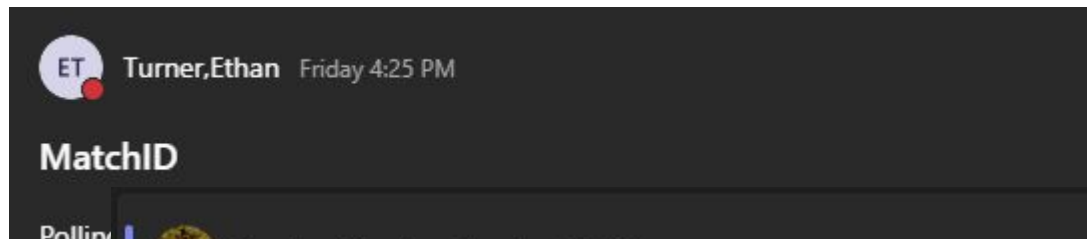
Agile

- Story Points
- P10 Sprint

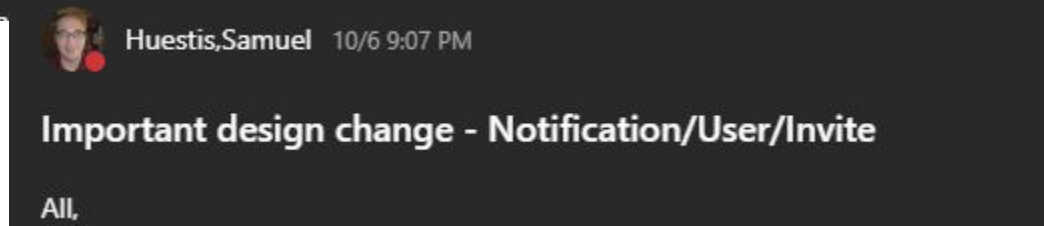
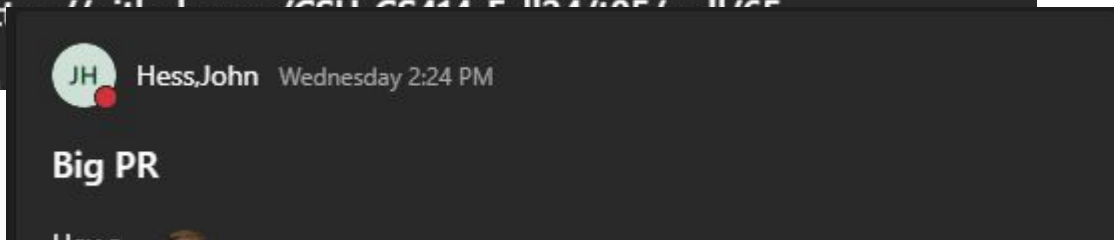


Agile Changes

- **More meetings!**
 - Bi-weekly most of the time
- **Messages on Teams**
 - Used teams to communicate more freely



Changes to Piece, creation of piece factory, new classes
[https://github.com/1001001001/1001001001/1001001001](#)

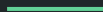


Important design change - Notification/User/Invite

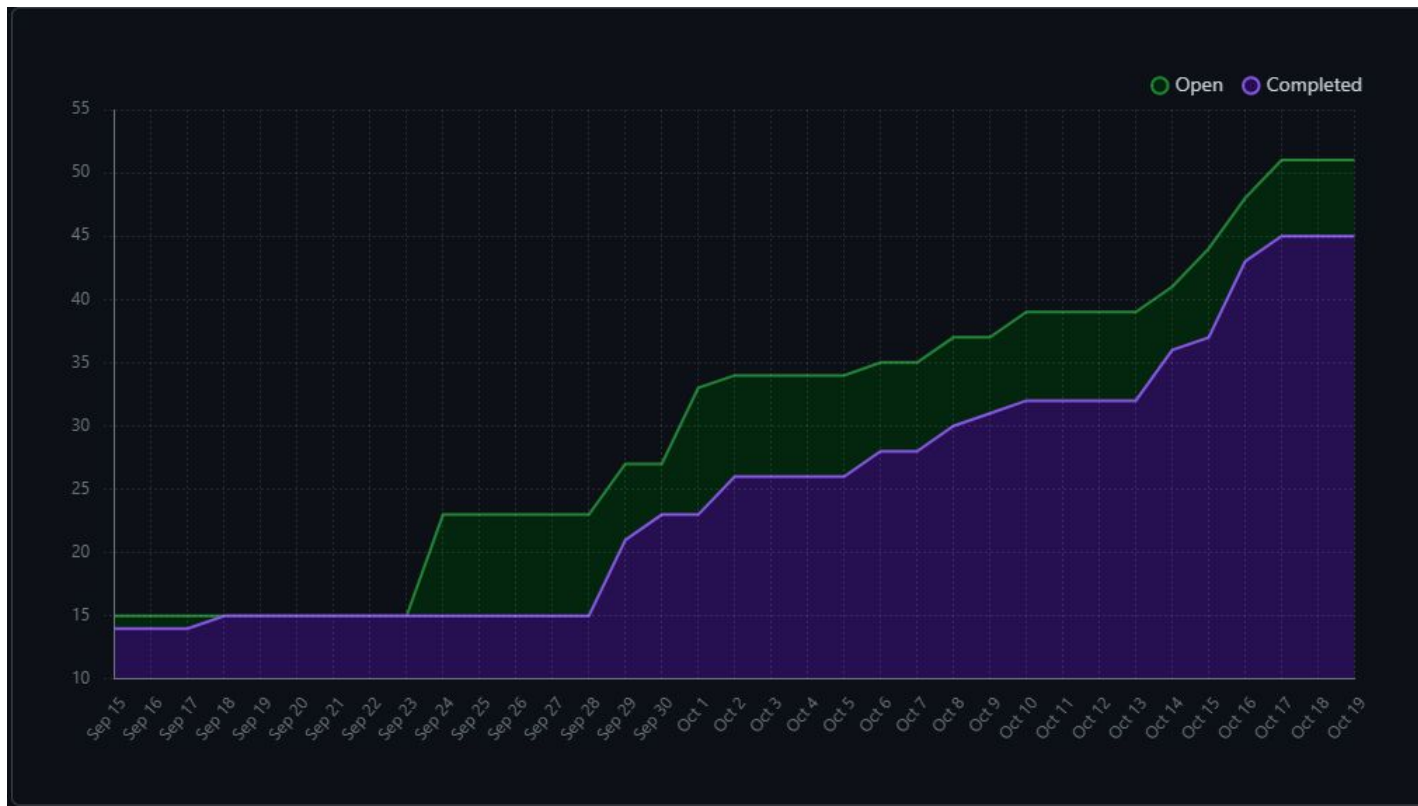
All,

Project Management

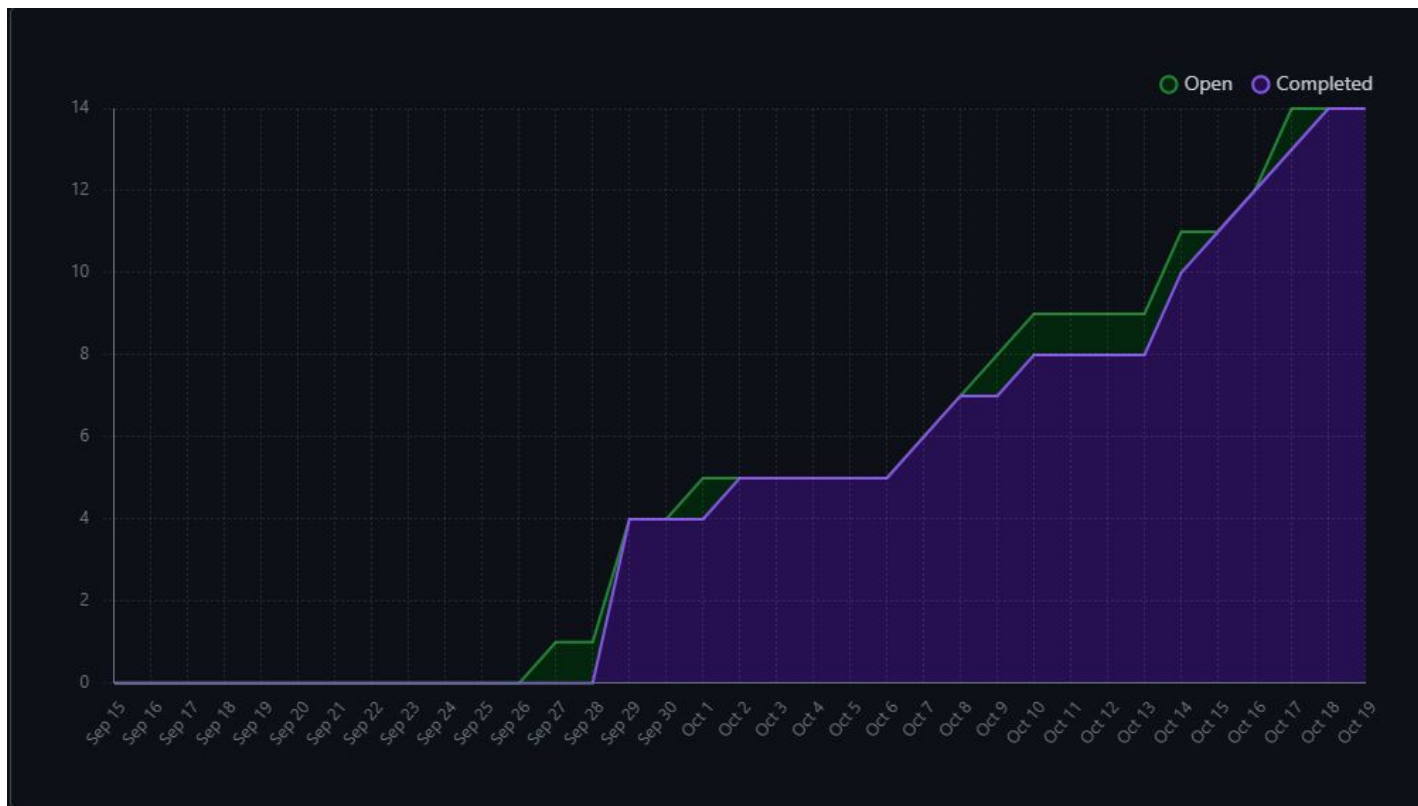
- Pull Requests
- Burnup Chart



Burnup Chart



Pull Requests Chart





QUESTIONS?

